

EXPERIMENT 5

Vacuum Cleaner Problem

Question

Write a Python program to solve the Vacuum Cleaner problem

Aim

To write a Python program to implement a **Vacuum Cleaner intelligent agent** that identifies and cleans dirty rooms.

Algorithm

1. Start with two rooms, A and B.
2. Check the current room.
3. If the room is dirty, clean the room.
4. If the room is clean, move to the other room.
5. Check the condition of the other room.
6. Clean the room if it is dirty.
7. Continue until both rooms are clean.
8. Display the final status of the rooms.

Code

```
python exp5.py - C:/Users/royal/python exp5.py (3.13.14)
File Edit Format Run Options Window Help
room = {
    "A": "Dirty",
    "B": "Dirty"
}

position = "A"

print("Initial State:")
print(room)

while room["A"] == "Dirty" or room["B"] == "Dirty":

    print("Vacuum is at", position)

    if room[position] == "Dirty":
        print("Cleaning", position)
        room[position] = "Clean"

    elif position == "A":
        print("Moving to B")
        position = "B"

    else:
        print("Moving to A")
        position = "A"

print("Final State:")
print(room)
print("All rooms are clean")
|
```

Output

```
===== RESTART: C:/Users/royal/Python/expo.py =====  
Initial State:  
{'A': 'Dirty', 'B': 'Dirty'}  
Vacuum is at A  
Cleaning A  
Vacuum is at A  
Moving to B  
Vacuum is at B  
Cleaning B  
Final State:  
{'A': 'Clean', 'B': 'Clean'}  
All rooms are clean  
|
```

Result

Thus, the **Vacuum Cleaner problem was successfully implemented using an intelligent agent**, and all dirty rooms were cleaned.